

FUNDAMENTALS OF COSTING AND PRICING

DAY 1



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LESSON TWO

COSTING METHODOLOGIES AND
PROCESSES

COSTING DIRECT MATERIAL USAGE -
CALCULATING RATINGS

LESSON TWO:

COSTING METHODOLOGIES AND PROCESSES:

Costing Methodologies

- Different types of companies & sectors use different costing methods. A **job costing method** is used when costs can be traced to a specific job such as building a house, catering for an event, doing an environmental impact assessment (EIA) or designing and printing an annual report. This is the most common method, especially for SMEs.
- But imagine a company mass-producing corn flakes, or facial tissues. Here it would be very difficult to trace exact costs to an individual box of corn flakes or a box of facial tissues. This company would most probably rather use **process costing**, an accounting system that calculates costs involved along a specific production process. This is usually the accepted method for costing mass produced products or large production chains.
- In this workshop, and because it is the more common solution in the SME and SMME space, we will focus primarily on job costing as a methodology. If you want to learn more about **process costing**, you can visit <http://www.accountingtools.com/overview-process-costing>

Costing Processes

- Raw materials are costed by weight, by number (quantity) or a combination of both.
- Sometimes when costing materials, we need to estimate usage for items that are difficult to quantify or measure, such as glue, paint, sewing cotton, certain food ingredients etc... this is essentially an educated guess, and we call this an **allowance**. It will usually be expressed as a Rand figure eg R2.00.
- You need to use excel or a similar programme, or at the very least a physical costing sheet and filing system to record all completed product & service costings for future reference. This will save you having to recost things all the time or do repeat costings for very similar products or services.
- Remember to **update your costs** should your suppliers raise their prices! Also, always try and keep general manufacturing costs as low as possible by minimising wastage. The concept of **circularity**, looks at ways of eliminating waste in your processes.
- Make sure all materials included in your costing are available when you need them. Having to substitute materials can seriously impact costs, delivery deadlines, profit & competitiveness. If this is a potential risk, build it into your quote.

Remember:

- Materials can be costed by weight, by quantity or a combination of both
- We sometimes need to estimate use and this is called an allowance
- Do not offer products or services that utilise materials and elements that may be hard to get in bulk if you receive a large order!

COSTING DIRECT MATERIAL USAGE - CALCULATING RATINGS

A rating is the quantity of a **particular raw material** that goes into an item. A rating tells us exactly how much wood, or sand, or fabric, or soap or even flour or curry powder we need to make a certain product, and then helps us easily work out what that raw material actually costs us, **per product**.

Ratings are best calculated using a Four Step process:

STEP ONE **Step One** is to decide **how you measure that particular material** that you are costing. Is it easiest to measure it by kilograms, or by the metre, or by litre, or by piece? Always choose the one that reflects *how you buy the material* in question.

STEP TWO **Step Two** is to assign a cost to that unit of measurement eg R50/kg or R100/m. We suggest you **round off** to make things easier – even if you buy 50 kg or 100 metres at a time – rather work out the cost for 1 kg or 1 metre, it is almost always simpler and more efficient from a costing perspective.

STEP THREE **Step Three** is to work out how much of each material you need to make a product. The most important thing, is to know **how many of that product** can be made from that metre of material, or out of that kilogram of flour, or from a square metre of aluminium etc.

This may entail laying out the pattern on a metre of fabric and estimating, or actually making up some samples and recording exact usage. Either way, it is key that you know **how many items** you can make from that 1 kg of flour, or 1 square metre of wood (based on the unit of measurement you chose in Step One).

Once you have worked this out, you can calculate **your rating** as follows:

If the item uses LESS THAN one unit of measurement i.e. less than a metre, kg etc. we apply the following formula:

$$1 \div \text{how many we can produce from that unit of measurement} = \text{our rating}$$

Conversely, if the item you are making uses **MORE THAN** one unit of the material we are measuring, your rating is simply **how much it uses**. It is crucial that you understand when to use the above formula, and when to simply capture how many units of measurement the item uses.

So, if you need 6 buttons to make a shirt, your rating is simply 6. If you need one zip, your rating is 1. If you need 25m of leather to make a lounge suite, your rating for leather is 25 - and so on.

NB: you only ever need to use the above formula if you can make more than one product from a unit of measurement (one metre of fabric, one kg of sand etc). If you cannot, your rating is simply how much (or how many) units of measurement you need to make the product e.g. 20kg of clay – rating is 20; 3m of fabric, rating is 3.

STEP FOUR THE LAST STEP is to work out **actual cost of that particular material** when making the product being costed.

To do this, we simply multiply our rating by the **cost per metre/kg/unit** of the material to get our answer:

Example:

$$\text{Rating} \times \text{Cost of purchase unit (kg/m etc)} = \text{Cost price per component per item}$$

If denim costs us R30/metre, and we know that we can make 4 denim caps from one metre, we would work out our rating for denim PER CAP as follows:

$1 \div (\text{how many we can make from one metre}) (4) = 0.25$. Therefore, our **rating** for denim when making this particular cap is 0.25.

To work out **the cost** of denim per cap, we multiply $0.25 \times \text{R30/m} = \text{R7.50}$

- If you can make 2 large hats from a metre, your rating will be $1 \div 2 = 0.5$.
- If you use 6 buttons, your rating is simply 6
- If you use 200g of flour to make a cake, this means you can make 5 cakes from 1kg so your rating is $1 \div 5 (200\text{g} \times 5 = 1\text{kg}) = 0.2$
- If you use 7kg of sand to make a paving slab, your rating is simply 7
- If you use 30 training manuals per workshop, your rating is simply 30

IMPORTANT: If the item uses MORE THAN our unit of measurement, we simply write down how many units we need, and that is our rating!

*So if we use LESS THAN our unit of measurement to make a product, we apply our formula - **divide 1 by how many we can make** to get our rating; if we use MORE THAN our unit of measurement (1kg, 1m) we **simply write down how many we will use** and that is our rating.*

*You then multiply your rating by the cost per unit of measurement, and that is your material cost for that component, done and dusted. **Easy??***

NOTES:

- When working out ratings, always make it as simple as possible for yourself by:
 - First deciding how you will measure that particular component (by the kg, metre etc)
 - Next, assign a cost to the unit of measurement you have selected, rounding it off to ONE wherever possible eg R30 for one kg
 - Third, establish how many products you can make from that unit of measurement: eg I can make 2 smoothies from one litre of yoghurt, OR I need 3 metres of denim to make one tablecloth
 - If you can make more than one item from a unit of measurement (eg 1kg), you apply the formula to get your rating:
- **$1 \div \text{how many we can produce from that unit of measurement} = \text{our rating}$**
 - If you use more than your unit of measurement, you simply write down how many/much you need, and that is your rating eg If I use 30l of oil to service an hydraulic pump, my rating for oil is 30.

Now you try a few:

MATERIAL COMPONENT AND COST PER UNIT OF MEASUREMENT	HOW MANY OF THE PRODUCT YOU CAN MAKE FROM ONE UNIT OF MEASUREMENT	YOUR RATING	COMPONENT COST PER PRODUCT
E.g. Galvanised wire at R35/kg	6	$1 \div 6 = \mathbf{0.167}$	$0.167 \times R35 = \mathbf{R5.83}$
Flour at R40/kg	20		
Linen at R60/metre	5		
Rose Geranium at R1000/l	500		
Plastic sheeting at R80/sqm	4		
Beads at R20/500g (R40/kg) (hint: first work out what a kg of beads costs – and double how many you can make)	2 (4 from a kg)		

